

Pixel Art — Count the Boxes (Beginner)

Builder Basics · Warm-Up · about 20 minutes

We color red boxes on a wall. First we count. 

 Do this before Week 1.

1 · X and Y 🧐

Open the **Pixel Art** world. Look at the wall.

Numbers go two ways:

- **X** = across →
- **Y** = up ↑

Y \ X	0	1	2	3
2				
1				
0				

Top row = **X**. Side = **Y**.

2 · One Red Box

Find the red box. Count to it.

Y \ X	0	1	2	3
2				
1				
0				

Across → 0, 1, **2**. **X = 2**

Up ↑ 0, **1**. **Y = 1**

START = (2, 1)

END = (2, 1)

One box. START and END are the **same!** 

This is the code. Two ways:

Blocks

```
fill   
  from X 2 Y 1 Z 0  
  to   X 2 Y 1 Z 0
```

Python

```
blocks.fill(RED_WOOL, world(2, 1, 0), world(2, 1, 0))
```

See? **2 1 0** and **2 1 0**. Same!

Write the box: X = __ Y = __

3 · A Red Line ■

A line has two ends.

 = START ·  = END

Y \ X	0	1	2	3	4
1					
0					

Across → START at **X 1**. END at **X 3**.

Up ↑ Both on **Y 1**.

START = (1, 1)

END = (3, 1)

Flat line. The **Y is the same!** (1 and 1)

Write it: START = (__ , __) END = (__ , __)

4 · Your Turn

Count the red box. Write the numbers.

Y \ X	0	1	2	3
2				
1				
0				

X = __ Y = __

One box. Are START and END the same? yes · no

5 · Build the Picture 📷

Go to your world. Build this red picture on the wall. ■

Y \ X	0	1	2	3	4	5	6	7	8	9	10
10	■	■	■	■	■	■	■	■	■	■	■
9	■	■	■	■	■	■	■	■	■	■	■
8	■	■	■	■	■	■	■	■	■	■	■
7	■	■	■	■	■	■	■	■	■	■	■
6	■	■	■	■	■	■	■	■	■	■	■
5	■	■	■	■	■	■	■	■	■	■	■
4	■	■	■	■	■	■	■	■	■	■	■
3	■	■	■	■	■	■	■	■	■	■	■
2	■	■	■	■	■	■	■	■	■	■	■
1	■	■	■	■	■	■	■	■	■	■	■
0	■	■	■	■	■	■	■	■	■	■	■

Count to each red box. Take your time. Do the boxes you can.

Take a photo or a video. Say:

My box is at X ... , Y ...

One box has START and END the same.

Submit ✓

Send to teacher on KakaoTalk.